

## Problem-Solution Fit

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|---------------|---------------------------------|
| Date          | 26 June 2026                    |
| Team ID       |                                 |
| Project Name  | Credit Card Approval Prediction |
| Maximum Marks | 5 Marks                         |

### Problem-Solution Fit Canvas:

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| <b>1. CUSTOMER SEGMENT(S) [CS]</b> <ul style="list-style-type: none"> <li>Financial institutions and banking firms</li> <li>Loan officers and credit analysts</li> <li>Individuals seeking a new credit card</li> </ul>                                                                                                                                                                                                                                                             | <b>6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]</b> <ul style="list-style-type: none"> <li>Applicants often have limited financial disclosures</li> <li>Application details may be incomplete</li> <li>Income sources vary widely</li> <li>Credit history on file may be minimal</li> <li>Heavy reliance on manual review steps</li> </ul>                                                                          | <b>5. AVAILABLE SOLUTIONS PROS &amp; CONS [AS]</b> <ul style="list-style-type: none"> <li>Manual checks: reliable but slow and expensive</li> <li>Rule-based systems: quick but not adaptable</li> <li>ML-based models: fast, scalable, accurate, and improve with historical data</li> </ul>                                                 |
| <b>2. PROBLEMS / PAINS + ITS FREQUENCY [PR]</b> <ul style="list-style-type: none"> <li>Manual review takes considerable time</li> <li>Evaluation errors happen due to human involvement</li> <li>High daily volume of incoming applications</li> <li>Approval decisions are delayed and inconsistent</li> </ul>                                                                                                                                                                     | <b>9. PROBLEM ROOT / CAUSE [RC]</b> <ul style="list-style-type: none"> <li>Application volumes keep growing</li> <li>Verification is still largely manual</li> <li>Human-driven decisions vary in consistency</li> <li>Applicant information is often incomplete</li> <li>Risk evaluation lacks efficiency</li> </ul>                                                                                                   | <b>7. BEHAVIOR + ITS INTENSITY [BE]</b> <ul style="list-style-type: none"> <li>Applicants want a fast turnaround on decisions</li> <li>Banks handle a very high daily application count</li> <li>There's a clear need for quick, consistent, dependable evaluation</li> </ul>                                                                 |
| <b>3. TRIGGERS TO ACT [TR]</b> <ul style="list-style-type: none"> <li>A rise in the number of credit card applications</li> <li>Growing demand for faster approvals</li> <li>Customers expecting instant decisions</li> <li>Pressure to cut down operating costs</li> </ul> <b>4. EMOTIONS BEFORE / AFTER [EM]</b><br><b>Before: Uncertainty, frustration, prolonged waits, and stress</b><br><b>After: Quicker approvals, greater confidence, transparency, and improved trust</b> | <b>10. YOUR SOLUTION [SL]</b><br>Build a Credit Card Approval Prediction System that leverages Machine Learning (Logistic Regression, Decision Tree, Random Forest, and XGBoost) integrated into a Flask-based web application.<br><br>The system instantly predicts approval or rejection using applicant details, improving accuracy, cutting down processing time, and helping banks make more dependable decisions. | <b>8. CHANNELS OF BEHAVIOR [CH]</b><br><b>ONLINE</b> <ul style="list-style-type: none"> <li>Banking websites</li> <li>Mobile banking apps</li> <li>Online credit card portals</li> </ul> <b>OFFLINE</b> <ul style="list-style-type: none"> <li>Bank branch locations</li> <li>Customer support centers</li> <li>Financial advisors</li> </ul> |